

Dharik Purohit

Portfolio: www.dharik.ca | LinkedIn: [linkedin.com/in/dharik-purohit-101](https://www.linkedin.com/in/dharik-purohit-101)
Montreal, QC | dharikp@gmail.com | 514-243-7600

Multidisciplinary engineer with hands-on experience in robotics system integration, mechanical design, instrumentation, and embedded development. Proven leadership in developing autonomous ground vehicles, sensor-driven systems, and electro-mechanical components. Adept in CAD, prototyping, ROS2, C++, Python, and electronics troubleshooting.

- ❑ **Robotics System Integration:** Led the mechanical development and system integration of a Mars rover at Space Concordia, including the design and fabrication of a 6-DOF robotic arm and a rocker-bogie suspension system, integrating sensor and actuator subsystems using ROS2 and embedded controllers.
 - ❑ **Embedded Systems & Instrumentation:** Designed a low-cost, high-precision potentiostat as part of a master's thesis, including custom PCB design, analog signal processing, and embedded troubleshooting—reducing costs by over 95% while maintaining sub-nanoampere measurement accuracy.
 - ❑ **Mechanical Design & Prototyping:** Executed full-cycle mechanical design and prototyping across multiple projects, including an award-winning anaerobic digester and rover components, utilizing CAD (SolidWorks, CATIA), CFD, 3D printing, CNC machining, and sheet metal fabrication.
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CORE COMPETENCIES

- **CAD & Design Software:** SolidWorks, Fusion 360, CATIA, AutoCAD, NX
 - **Prototyping & Fabrication:** CNC machining, manual lathe and milling operations, 3D printing, sheet metal fabrication, process sheets
 - **Embedded & Instrumentation:** Altium, Basic sensor integration, PCB designing
 - **Analysis Tools:** CFD and FEA in SolidWorks, CATIA, ANSYS
 - **Programming:** C++, C#, Python, ROS2, MATLAB, Simulink, Arduino, GitHub, ubuntu
 - **Languages:** Fluent in English (Written & Spoken) and intermediate in French (Written & Spoken)
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PROJECT EXPERIENCE

Mechanical Lead, Robotics Division
Space Concordia, Concordia University

05/2019 -04/2025

Led a 25+ person team developing Mars rover systems for University Rover Challenge (URC) and European Rover Challenge (ERC). Managed system documentation, design reviews, and mechanical integration timelines.

Key Contributions

- ❑ Designed and fabricated a custom 6-DOF robotic arm using cycloidal drives for high-torque applications.
- ❑ Developed and tested a rocker-bogie suspension to improve rover terrain adaptability.
- ❑ Engineered an aluminum electronics enclosure, incorporating CFD for thermal optimization.
- ❑ Integrated embedded systems with sensors, actuators, and control logic using ROS2 and Arduino.

Mechanical Engineering Master's Thesis
Concordia University

01/2023 - 04/2025

Designed a compact, low-cost potentiostat for electrochemical and liquid battery research. Aimed at increasing accessibility for smaller labs and students while supporting advanced features.

- ❑ Rapidly prototyped the full system, electronics, software, and mechanical design from concept to device.
- ❑ Designed and fabricated a PCB enabling electrochemical detection below 1 nA, while reducing the cost from \$10,000 to under \$200.
- ❑ Created mechanical housings and integrated the device with lab equipment, ensuring durability and usability in real-world applications.
- ❑ Iteratively debugged hardware, firmware, and instrumentation interfaces to deliver a test-ready system.

Capstone Engineering Design Project**09/2021 – 07/2022**

Concordia University

Designed and built an anaerobic digester for biogas generation in suburban homes capable of converting organic waste into biogas (methane) for energy applications.

Key Contributions

- ❑ Developed control systems for gas flow and combustion optimization.
- ❑ Actively involved in testing and validation processes.
- ❑ Created Design for Manufacturing (DMF), refining component layouts and material choices for cost-effectiveness, reliability and scalability.
- ❑ The project was awarded the Capstone Design Award for mechanical innovation.

PROFESSIONAL EXPERIENCE

Laboratory Assistant – Machine Drawing & Design**09/2023 – 12/2024**

Concordia University

- ❑ Instructed undergraduates on mechanical assembly/disassembly and CAD best practices.
- ❑ Evaluated technical drawings for accuracy and compliance with engineering standards.
- ❑ Provided one-on-one mentoring on mechanical systems and engineering documentation.

Flight Performance Intern**01/2021 – 04/2021**

Bombardier Inc., Montreal

- ❑ Debugged C# applications for flight data analysis and automation tools.
- ❑ Automated archival processes for aircraft telemetry data, reducing manual workload.
- ❑ Analyzed aerodynamic and flight control performance data to inform GUI tool updates.

R&D Intern – Space Division**08/2019 – 12/2019**

MPB Communications Inc., Pointe-Claire

- ❑ Designed MATLAB tools to process >15 million wavelength bands from spectrometry systems.
- ❑ Contributed to development of optical technologies for Mars rover prototypes.
- ❑ Validated performance of laser-based spectrometry hardware under test simulations.

EDUCATION

Master of Science, Mechanical Engineering, (GPA 3.6/4.3) Concordia University (2025)

Bachelor of Engineering, Mechanical Engineering (GPA 3.4/4.3), Concordia University (2022)